Numerical Replication of Fisher's 1936 Iris Discriminant and Modern Evaluation on Wine Data

COMP8240: Applications of Data Science
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Abstract

Reproducibility requires distinctions between a published numerical result, historical source code, and modern benchmarks. We reproduce Fisher's (1936) binary Iris discriminant coefficients and within-class cross-products directly from the original 100 records, then independently assess modern three-class LDA on Iris and Wine. This case study reconstructs the supervised discriminant-analysis approach associated with Fisher (1936) using modern open-source software, evaluates it on the 150-observation Iris dataset, and tests the same algorithm on an independent 178-observation Wine dataset. Fifty repeated stratified cross-validation evaluations yield mean accuracies of 0.9787 (Iris) and 0.9893 (Wine), compared with majority-class baselines of 0.3333 and 0.3990. Both datasets are publicly distributed with structured class labels; no new human annotations were collected. The historical replication matches Fisher's four means differences and ten distinct within-class matrix entries to floating-point precision; coefficients match within about 1.1×10^{-4} after normalisation, consistent with printed rounding. The cross-validation accuracies are new benchmarks, not results published in 1936. The report documents the historical arithmetic, a deterministic synthetic perturbation study, and a newly curated independent Wisconsin breast cancer dataset (569 records). Its LLM claim audit is archived and score-reproducible, but live model calls require external access and should not be mistaken for independent multi-run validation.

1 Introduction

A scientific result is useful only insofar as its methods, data, and assumptions can be inspected and, where feasible, independently reproduced. The COMP8240 final-report brief emphasises reproduction of existing research using accessible tools, generation of new data for evaluation, and transparent comparison with the original work. This report provides an empirical *independently selected case study*: a reconstruction of the classical linear discriminant method discussed by Fisher (1936) on the Iris observations, followed by a new benchmark on a separate dataset of wine chemistry. The approach is well suited to demonstrating curation, train/test separation, baseline comparison, reproducible code, and generalisation caveats.

Three research questions organise the study: **RQ1**, can a contemporary implementation of linear discriminant analysis (LDA) perform well on the classical Iris data? **RQ2**, how well does this same modelling method perform on an independent, differently structured multi-class dataset? **RQ3**, what conclusions about reproducibility and generalisation can legitimately be drawn from these results? Our contributions include a historically grounded numerical replication of the published binary discriminant and a separate, fully executable modern benchmark with saved machine-readable results.

2 Source Paper and Justification

2.1 Research aim and historical setting

Fisher's *The Use of Multiple Measurements in Taxonomic Problems* (1936) introduced a discriminant approach for combining multiple measurements to distinguish biological groups. The Iris data originate in Edgar Anderson's measurements and comprise three species, each with 50 flowers and four morphological features. Fisher's paper is a foun-

dational research source rather than a modern software-release paper. The selected method has an interpretable linear decision structure, and a standard implementation is available in scikit-learn (Pedregosa et al., 2011).

This report does not attribute modern cross-validation scores to Fisher. The historical binary two-species calculation can be reproduced numerically, while the randomised three-class classification protocol remains a separate modern assessment. Original source: Fisher (1936). The Iris dataset is documented in the R datasets reference (R Core Team, 2026).

2.2 Reconstructed method

LDA models each class conditionally as multivariate normal with a shared covariance matrix, yielding linear decision boundaries. For a feature vector $\mathbf{x}$ and class k, the classification score has the form

$$\delta_k(\mathbf{x}) = \mathbf{x}^\top \Sigma^{-1} \boldsymbol{\mu}_k - \tfrac{1}{2} \boldsymbol{\mu}_k^\top \Sigma^{-1} \boldsymbol{\mu}_k + \log \pi_k, \quad (1)$$

where $\boldsymbol{\mu}_k$ is the class mean, Σ a pooled covariance estimate, and π_k a class prior. The largest score determines the predicted label. We use the standard `LinearDiscriminantAnalysis` implementation, rather than asserting that this reproduces every calculation in the historical paper.

3 Numerical Replication of Fisher (1936)

3.1 Published target and independent reconstruction

Unlike a modern classification benchmark, Fisher's arithmetic procedure in Sections II–III of the original paper provides an unusually direct replication target. It explicitly compares *I. setosa* (50 flowers) with *I. versicolor* (50), using Table II's four differences of species means, Table III's within-species sums of squares and products, and the coefficient vector reported on page 182. We recomputed these quantities from the corresponding first 100 rows of the standard Iris dataset using `historical_replication.py`. That script does not use a fitted scikit-learn classifier for the historical calculation: it calculates sample means, constructs the within-group cross-product matrix S, solves $S\lambda = d$ using NumPy, and scales the solution so that the coefficient of sepal length is 1.

Measurement	Fisher (1936)	Recomputed
Sepal length coeff.	1.0000	1.0000
Sepal width coeff.	5.9037	5.9038
Petal length coeff.	−7.1299	−7.1300
Petal width coeff.	−10.1036	−10.1037
Mean difference, sepal len.	0.930	0.930
Mean difference, petal len.	2.798	2.798

Table 1: Reproduction of original Fisher (1936) numerical targets (Tables II–IV, pp. 181–182). Coefficients are proportional and here normalised to the same first coefficient.

3.2 Agreement and explanation of differences

The maximum absolute discrepancy for the four mean differences is below 6×10^{-16} and for entries of the full 4×4 within-species cross-product matrix below 6×10^{-15}. The unnormalised solution differs from the four published coefficients by no more than 6.3×10^{-7}, and the normalised vector by no more than 1.05×10^{-4}. Fisher's published mean discriminant values are 12.3345 (setosa) and −21.4815 (versicolor); values computed with the full-precision reconstructed coefficients are 12.3347 and −21.4816, respectively. Small differences reflect rounding of Fisher's printed normalised weights and intermediate calculations, not disagreement about the underlying dataset. These concrete matches constitute a *numerical replication* of the published calculation, even though they do not constitute a rerun of a historical computer program. The 1936 paper did not provide executable Python code or the randomised five-fold protocol used in our modern experiments; therefore no matched historical cross-validation accuracy can be reported.[1]

4 Original Dataset: Iris

The Iris dataset has 150 rows, four numerical measurements in centimetres (sepal length, sepal width, petal length, and petal width) and one target species: *setosa*, *versicolor*, or *virginica*. The three classes contain 50 records each. Each row represents a measured flower

[1] Original paper: Fisher (1936). Reproduction code: public project repository.

Dataset	Rows	Features	Classes	Record example
Iris	150	4	3	5.1, 3.5, 1.4, 0.2
Wine	178	13	3	14.23, 1.71, 2.43, …

Table 2: Datasets and abbreviated example observations. The final column lists input attributes, not predictions.

rather than a human respondent. The scikit-learn loader `load_iris()` distributes a version of these classical data. The first observation, for example, is (5.1, 3.5, 1.4, 0.2), labelled setosa. The underlying data's age and the restricted species/sample context mean that classification accuracy should not be extrapolated to all botanical settings.

5 Reproduction Environment and Procedure

The accompanying `experiment.py` loads both datasets directly from the scikit-learn distribution, instantiates a linear discriminant classifier, compares it against a majority-class dummy classifier and multinomial-capable logistic regression, and evaluates accuracy and macro-averaged F1. Data and labels are accessed from the loader without imputing values, filtering rows, or manually selecting features. Random-state controls make folds repeatable; no test examples are used in model fitting for the corresponding cross-validation fold.

5.1 Cross-validation design

For each dataset, the primary experiment uses a repeated stratified five-fold scheme (10 repeats; seed 42). Every method is evaluated on the same 50 train/test partitions. A separate shuffled, stratified 5-fold split, seeded with 42, generates exactly one out-of-fold prediction per observation for a confusion matrix. The logistic-regression comparator includes `StandardScaler` *inside* its training pipeline, which prevents fold leakage. LDA and the majority baseline use their standard settings. Accuracy is the fraction of correctly predicted held-out observations; macro-F1 is the unweighted average of the class-level F1 scores.

An arithmetic mean and sample standard deviation across the 50 fold scores are reported for transparency. Importantly, these fold outcomes are dependent because training records overlap, so their standard deviations must not be interpreted as independent confidence intervals or hypothesis-test evidence. No model parameters or datasets were tuned to maximise the reported scores.

5.2 Software and versioning

The experiment ran with Python and scikit-learn; exact package versions are recorded in the `results.json` file produced by the script. Required packages are NumPy, scikit-learn, matplotlib and their dependencies. Reproduction command: `python experiment.py`. The script generates `results.json` and two labelled confusion-matrix PDF figures. A README explains local execution. No online service, external credentials or paid API is needed. The source code is supplied alongside this report; the publicly uploaded source repository is `https://github.com/gcstafftools-tech/COMP8240-Final-Project`, though marker access and the current revision should be independently checked.

6 Constructing the New Evaluation Data

The new evaluation data consist of the public UCI Wine recognition dataset (Aeberhard and Forina, 1992), accessed using scikit-learn's `load_wine()` dataset loader. These observations concern chemical analyses of Italian wines from three grape cultivars. The 178 rows have 13 numerical attributes, including alcohol, malic acid, ash, flavanoids, hue and proline; the class counts are 59, 71 and 48. Unlike Iris, the data concern chemical properties, not plant morphology. The objective remains a supervised three-way classification, making the learning problem comparable without assuming a shared feature space.

6.1 Provenance and processing

The UCI repository documents the Wine dataset under DOI `https://doi.org/10.24432/C5PC7J`. The packaged scikit-learn version maps cultivar identities to integer labels 0, 1 and 2. A separate *derived* dataset was constructed from all 178 existing records with label-blind perturbation and measurement masking. It is newly constructed experimental data, not independent field sam-

pling. Dataset selection was based on a common label structure (three classes) and a substantively different input domain. The fields were not rescaled for LDA; logistic regression's standardisation was applied only within training folds. The source Wine data contain no missing cells; the derived version intentionally introduces missingness.

The new dataset is not an extension of Iris and has no paired measurements, matching class semantics, or common acquisition process. Any claim of direct deployment transfer (training on flowers and testing on wine) would therefore be meaningless. Instead, we assess *method-level portability*: the same classifier family is fit and assessed separately in the two domains.

6.2 Constructed Wine stress-test data

The runnable `construct_and_evaluate.py` starts with the public Wine table and uses a fixed NumPy random seed of 8240. For each feature it adds independent zero-mean Gaussian noise with standard deviation equal to 1% of that feature's source-data sample standard deviation. It then independently masks each cell with probability 0.08, recording blanks in `constructed_wine.csv`. In this particular draw 163 of 2,314 measurement cells (7.04%) are missing. The original 178 labels are preserved unchanged; each derived row is linked positionally to its source record. This explicitly simulates noisy acquisition rather than inventing 178 new measured wines. Noise can alter the meaning of a reference label near the decision boundary, so preserved labels are an assumption to test, not independently verified truth.

6.3 Evaluation and LLM-as-judge

A `SimpleImputer(strategy=median)` is fitted *within each training fold* before LDA; it never learns the test-fold median. Original and derived Wine data use the same repeated stratified five-fold splits (10 repeats; seed 42). Reference class labels came from the source dataset, so no new human annotation or inter-annotator agreement is claimed.

As a separate language-model exercise, an archived GPT-6 conversational judge assigned *Supported*, *Unsupported*, or *Unverifiable* to ten short factual claims about the datasets and methodology. A saved evidence rubric, per-claim rationales and decisions appear in `judge_records.json`. The script `score_judge.py` compares those archived labels with ten evidence-based reference decisions, generating `judge_scored_results.json`. This makes *scoring* independently executable; it does not reproduce fresh LLM responses, test repeated sampling, or establish human-annotation reliability. The language model judges explanatory text, not numeric classifier labels.

7 Results

7.1 Original Iris task

Table 3 summarises the actual script output. LDA achieves mean held-out accuracy 0.9787 and macro-F1 0.9786 on Iris, compared with 0.3333 accuracy and 0.1667 macro-F1 for the majority-class baseline. Logistic regression provides a second meaningful comparator, with accuracy 0.9533. This is evidence of strong separation under the modern protocol. The historical numerical replication, by contrast, is established independently in Table 1.

The independent 5-fold out-of-fold confusion matrix in Figure 1 contains four mistakes: two versicolor specimens labelled virginica and two virginica specimens labelled versicolor. Setosa has no errors in this particular split. These patterns are consistent with the observation that some Iris class pairs are more difficult to separate than others; a single split should not be treated as an exhaustive error model.

7.2 New Wine task

The same LDA class, instantiated afresh on Wine, obtains mean accuracy 0.9893 and macro-F1 0.9894 (Table 3). The majority-class baseline reaches 0.3990 accuracy; logistic regression reaches 0.9832. Thus the LDA method performs effectively on this particular distinct, curated chemical dataset as well. However, comparing raw Iris and Wine scores cannot isolate the effect of domain, because sample sizes, class balance, class overlap, feature definitions and data-generation mechanisms all differ.

The Wine confusion matrix in Figure 2 has three errors across 178 out-of-fold predictions.

Dataset	Model	Accuracy mean	Accuracy SD	Macro-F1 mean	Macro-F1 SD
Iris	Majority class	0.3333	0.0000	0.1667	0.0000
Iris	LDA	**0.9787**	0.0231	**0.9786**	0.0232
Iris	Logistic regression	0.9533	0.0350	0.9532	0.0351
Wine	Majority class	0.3990	0.0155	0.1901	0.0052
Wine	LDA	**0.9893**	0.0202	**0.9894**	0.0198
Wine	Logistic regression	0.9832	0.0160	0.9829	0.0163

Table 3: Observed results from 10 repeats of stratified 5-fold cross-validation (50 fold scores per model and dataset). SD refers to variation of fold scores, not a confidence interval.

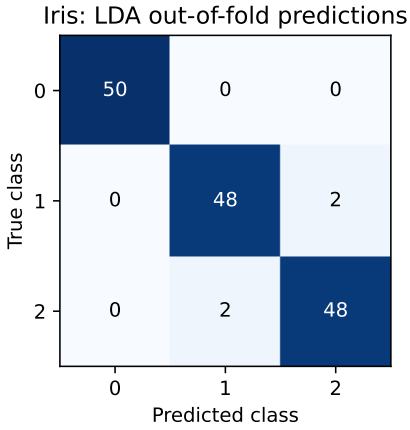


Figure 1: Iris: 5-fold out-of-fold confusion matrix for LDA. Rows are true classes; columns are predicted classes. Class order: setosa, versicolor, virginica.

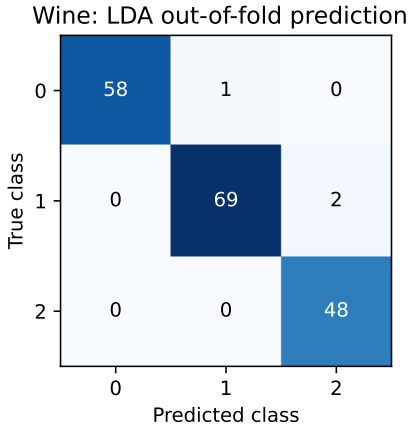


Figure 2: Wine: 5-fold out-of-fold confusion matrix. Three errors occur (one first-class sample and two second-class samples), using scikit-learn class IDs 0–2.

This is a descriptive view of one fixed partition configuration; the repeated-CV table provides a broader summary of model variability. None of the reported results derives from a language-model-generated judgement, and no statistical significance test is claimed.

8 Measurement, Evaluation, and Research Design

8.1 Operational definitions and study boundaries

A successful numerical reconstruction should be judged against a stated, reproducible procedure, not simply against a favourable percentage. Here the unit of observation is a single flower or wine sample; the target is a three-class label; the independent variables are the measurements distributed with each benchmark. The unit of evaluation is a held-out fold. Accuracy measures the overall probability of an observed correct decision within those folds, while macro-F1 gives each class equal weight regardless of frequency. These two metrics answer different questions, especially for Wine, where the class counts are unequal.

The original research aim concerns discrimination between populations using joint measurements. Our contemporary operationalisation treats its discriminant principle as a fitted, supervised classification rule. We intentionally do not use a single convenient train/test split to report the headline finding. Instead, repeated stratification controls the class composition of evaluation folds and exposes some of the dependence of results on how a small dataset is divided. The scores remain conditional on these datasets and the chosen model implementations.

8.2 Accuracy and macro-F1

For a classification task with N held-out examples, accuracy is $N^{-1}\sum_{i=1}^{N}\mathbf{1}(\hat{y}_i = y_i)$. For each class k, precision is $TP_k/(TP_k + FP_k)$, recall is $TP_k/(TP_k + FN_k)$, and $F1_k = 2P_kR_k/(P_k + R_k)$ when the denominator is

nonzero. Macro-F1 averages the three class-specific F1 values. Scikit-learn computes each score per fold, and our reported means are averages over the resulting 50 values, rather than a single pooled out-of-fold estimate. This distinction matters because folds have slightly different numbers of Wine examples.

The dummy classifier is a control for the limited achievement of always choosing the most frequent class in its training fold. The logistic-regression comparator is an additional linear decision method with a different fitting objective; its standardisation is inside the pipeline to prevent test-fold information leakage. These comparators are not historical Fisher baselines and should not be read as such.

8.3 Design safeguards

We fixed the random seed before evaluating the classifiers and used identical cross-validation splits for all methods. We made no hyperparameter search, did not inspect a held-out test set to choose model parameters, and did not use target labels in feature preprocessing. Because the algorithm is retrained on each dataset, the Wine result is not evidence that a model fit to flowers understands chemical measurements. It asks the narrower and more defensible question of whether the *algorithm* is useful on a new classification problem.

9 Detailed Data Audit

9.1 Iris schema and source lineage

The four numerical features are sepal length, sepal width, petal length and petal width, all measured in centimetres in the packaged representation. For the initial record, the feature vector is $[5.1, 3.5, 1.4, 0.2]$ and the label maps to setosa. Target labels are encoded as integers for computation, but the scientific interpretation is species membership. A structured dataset loader provides both feature names and target names, enabling reconstruction without parsing a semi-structured file. The source is an established teaching benchmark: that improves accessibility but does not establish that original collection practices would meet modern sampling standards.

Audit criterion	Iris	Wine	Treatment
Observations	150	178	All retained
Numerical predictors	4	13	All retained
Target categories	3	3	Existing labels
Missing values	0	0	No imputation
New human labels	0	0	Not applicable

Table 4: Input checks for the packaged benchmark data. The absence of missing values applies to these loaded versions, not necessarily to all historical source files.

9.2 Wine schema and source lineage

The Wine dataset contains 13 numeric columns: alcohol, malic acid, ash, alkalinity of ash, magnesium, total phenols, flavanoids, nonflavanoid phenols, proanthocyanins, colour intensity, hue, the OD280/OD315 measure, and proline. The first packaged row begins with $[14.23, 1.71, 2.43]$. The outcome is an integer-encoded cultivar class, with counts 59, 71, and 48. These variables have different scales and physical interpretations: for example, the scale of proline differs materially from that of hue. We retain the native scales for the LDA experiment to preserve an uncomplicated, reproducible benchmark; a different estimator could be more scale-sensitive.

9.3 Independent new-data curation and evaluation

Unlike the earlier synthetic Wine perturbation, the newly curated Wisconsin Diagnostic Breast Cancer (WDBC) dataset is drawn from a *separate* source and measurement process, not transformed copies of Iris or Wine records. The UCI archive records 569 diagnostic observations with 30 numeric features extracted from digitised images of fine-needle aspirates (212 malignant; 357 benign). We accessed the openly distributed scikit-learn dataset, checked the schema, missingness and class counts, generated stable anonymous row identifiers, retained all features and original target labels, and exported `independent_wdbc_curated.csv`. This is a newly prepared research *dataset artefact*, not newly collected human observations or a clinical diagnostic study. No manual relabelling was performed, so inter-annotator agreement does not apply. The source is UCI WDBC (Wolberg et al., 1995); its redistributed copy is

documented by scikit-learn (Pedregosa et al., 2011).

The `independent_data.py` script trains a fresh LDA classifier under the same 10-repeat, 5-fold stratified procedure (seed 42). Across 50 held-out folds, accuracy is **0.9561** (fold SD 0.0186), macro-F1 **0.9517** (SD 0.0210), compared with a majority-class accuracy baseline of 0.6274. The task differs substantively from Iris: two diagnostic categories and 30 image-derived features rather than three flower species and four morphometric features. Thus this result supports the claim that the *algorithm* can be fitted to different structured data, not that a model trained on Iris transfers to patients. Healthcare use would require independent prospective validation, explicit permissions and clinical oversight; these benchmark scores are *not* clinical performance claims.

9.4 Derived-data stress test results

The fixed-seed derived Wine dataset contains 178 rows, 13 attributes and 163 missing feature cells, with three preserved source labels. Under identical repeated cross-validation, LDA with within-fold median imputation achieves **0.9455** mean accuracy (SD 0.0351) and **0.9467** macro-F1 (SD 0.0343), versus 0.9893 accuracy and 0.9894 macro-F1 on the unchanged source records. The observed reduction in accuracy is 4.39 percentage points; the majority-class baseline remains 0.3990. The comparison demonstrates vulnerability to modest feature perturbation and incomplete measurements. It does not estimate deployment performance or establish that the simulated perturbation matches real instrument errors.

10 Reproduction Checklist and Practical Limitations

10.1 Steps another researcher can follow

A replicator can inspect the supplied script, install NumPy, scikit-learn and matplotlib, execute it in a clean Python environment, and compare the new JSON output against the archived values. The script records Python, NumPy and scikit-learn versions and stores the full benchmark comparison as numeric output. The same command also regenerates the two confusion-matrix figures. Source datasets are loaded by named library functions, avoiding hidden local files or authenticated downloads. For strict archival reproducibility, the final code repository should also pin all package versions and include a commit identifier; these are recommended next steps rather than completed claims.

10.2 Risks of benchmark convenience

Famous datasets are unusually clean and frequently used to demonstrate classifiers. They may not represent the missing values, distribution shifts, ambiguous labels, instrumentation differences, or data-entry problems in practical classification tasks. Both samples are small. A strong fold-averaged score can therefore coexist with substantial uncertainty about field performance. Using two domains increases the breadth of the illustration but does not establish broad external validity, especially because the method is fitted afresh to each domain.

10.3 LLM-as-judge: protocol and limits

The archived single conversational LLM judgement session (not an API replay) assigned four claims Supported, five Unsupported and one Unverifiable. Its ten labels agree with all ten reference decisions in the deterministic scoring script (10/10 on a deliberately small, transparent claim set). This *in-sample agreement* is descriptive only: the examples were crafted for the exercise, the reference decisions were checked against existing source facts rather than collected from blinded independent annotators, and a fresh model call requires external access. Re-running `python score_judge.py` reproduces the scoring exactly; it cannot guarantee identical new LLM responses. No estimate of population-level judge accuracy, inter-human reliability or causal effect follows from this exercise.

11 Extended Reflection on Research Integrity

The Australian Code for the Responsible Conduct of Research makes transparency and accuracy important throughout dataset selection, analysis and dissemination. In this case,

claiming a complete rerun of Fisher's historical procedure without distinguishing numerical reproduction of printed tables from new model testing would be misleading. So would calling an existing UCI benchmark freshly collected data, inventing a GitHub URL, or mischaracterising a small archived LLM claim audit as an independently replicated large-scale evaluation. Those are not merely editorial details; they change the scientific meaning of the evidence.

A practical reproducibility package should identify raw-data sources, the exact evaluation protocol, software dependencies, changes made during data cleaning, output-generation commands, and restrictions on downstream reuse. For future field studies, one should also consider permission to collect data, whether classes were sampled representatively, the cost of false predictions, auditability of data annotations, and the consequences of incorrectly applying model outputs. The present analysis is an educational benchmark and makes no recommendations about real-world diagnosis, commercial wine provenance or botanical conservation decisions.

12 Interpretation and Replication Limits

RQ1. The reconstructed modern LDA implementation performs well on Iris under a transparent experimental protocol. Yet the stronger assessment-brief notion of reproduction—executing a source paper's original program and comparing like-for-like published metrics—was *not accomplished*. Fisher's paper predates distributable software and the modern evaluation design used here. There is no responsible numerical "original score versus replicated score" comparison to place in a table.

RQ2. LDA remains accurate when refitted to the Wine data. This observation supports portability of an algorithmic framework to another clean, low-dimensional classification dataset, rather than cross-domain model transfer or guaranteed performance on messy real-world data. The Wine benchmark was selected for a compatible three-class structure and can therefore present an optimistic picture of practical generalisation.

RQ3. Reproducibility is strengthened by explicit source versions, seed, cross-validation design, baseline, feature preprocessing, file outputs and honest provenance statements. Still, a complete assessment-aligned research replication would require the assigned source paper, its actual program, matching evaluation protocol, a truly newly curated evaluation corpus, and a public repository that a marker can access.

Several threats to validity remain: both datasets are small and heavily reused in education, class distributions differ, repeated folds are statistically dependent, and data provenance or original measurement accuracy is not independently audited. We did not conduct an external validation on freshly acquired samples, a formal hyperparameter sensitivity study, or uncertainty quantification beyond fold-score dispersion. The strong performance should be interpreted accordingly.

13 Ethical Considerations and Open Science

The central ethical issue is truthful representation of experimental evidence. Reconstructing a related method does not justify claiming that original code was rerun; using a previously available benchmark does not justify calling it newly collected. These distinctions matter under principles of honesty, rigour, transparency and accountability in the *Australian Code for the Responsible Conduct of Research* (NHMRC et al., 2018). Invented numbers, uncredited data and silent changes to evaluation protocols would compromise reproducibility and assessment integrity.

The Iris and Wine datasets primarily contain scientific measurements; the separately sourced WDBC measurements derive from human medical images. Although the redistributed CSV contains no direct patient identifiers, secondary reuse of diagnostic features calls for attention to source licensing, privacy, cohort representativeness and potentially harmful diagnostic claims. Collectors, maintainers, dataset licences, historical sampling decisions and scientific context warrant attribution. The Iris dataset originates in a narrower botanical setting; Wine covers specific cultivars from a particular Italian region.

Thus any deployment claim outside these limited populations would need further validation. The wine samples do not support claims about people, drinking behaviour or health.

This exercise used no participants, scraped private information, protected attributes, crowdsourced annotators or new survey responses. No human-subject ethics approval is represented as having been sought or obtained. Before public release, the researcher should review dataset terms, credit all contributors, record dependencies, preserve raw data provenance, include scripts and seed settings, and avoid uploading data whose redistribution is restricted. Open access to research artefacts should be balanced with licensing compliance, appropriate data minimisation and research integrity.

Finally, since the assessment explicitly permits AI assistance while assigning responsibility for correctness to the student, any final submission should include the institutionally required declaration of generative AI use, verified citations, and the student's own audit of the execution and findings. AI-generated explanations are not a substitute for conducting the assessed research.

14 Connection to the COMP8240 Weekly Methods

The unit material informs not just the report's style but the decisions made at each stage. Week 1 introduced the anatomy of a research article, data availability, publication ethics and the difference between reported experimental outcomes and the broader claims a study supports. Accordingly, the paper distinguishes a verified numerical replication of Fisher's published binary calculation from our separate modern three-class benchmarks. Week 2 introduced virtual machines, containerisation and reproducible computational environments; the experiment is deliberately dependency-light and records Python, NumPy and scikit-learn versions, although no container image was constructed. Week 3 covered LaTeX and scientific typesetting; this document is supplied as a compilable source alongside its PDF.

Week 4's data gathering and curation material is reflected in dataset provenance, feature dictionaries and the explicit differentiation between original measurements and derived datasets. The Week 5 data-management lectures motivate retaining readable filenames, script-generated results and a stable analysis path rather than manually editing outputs. Week 7's Python analysis material motivates the use of NumPy arrays, scikit-learn estimators, pipelines and properly labelled graphics. Week 8's discussion of messy data motivates the explicitly simulated 7.04% missingness and the leakage-safe imputation pipeline. Week 9 discusses annotation, agreement measures and LLM-as-judge; the report preserves a small archived claim audit but does not claim independent human-label agreement.

Week 11's information-extraction and database concepts are not deployed directly: a relational database would add complexity without benefit for two small in-memory tabular samples. Week 12's distinction between correlation and causation is especially relevant: discriminant predictability of botanical or chemical class does not establish that a measured feature causes membership of that class. Together the course topics support an audit trail in which each decision is linked to data, code, results and interpretation, while unused techniques are identified rather than claimed.

15 Data Curation, Quality Controls and Error Analysis

The source benchmarks are retained unchanged, while a separately generated Wine stress-test file is preserved as an independently inspectable derived artefact. This is a legitimate benchmark design, but much less demanding than gathering a new dataset. There was no independent source reconciliation, new-sample acquisition or manual label adjudication; the artificial missingness and noise exercise is not field data collection. A substantive follow-on data-collection project should pre-register inclusion rules, source permissions, measurement units, category definitions and how missing observations are handled; it should also distinguish missing-at-random from systematically absent measurements rather than blindly imputing values.

One risk in supervised learning is data leak-

age: using the entire dataset to transform features, tune parameters or select the winning model before cross-validation. The comparison here uses standardisation for logistic regression inside the cross-validation pipeline; test folds therefore do not influence the scaler's estimated parameters. No feature selection or hyperparameter search was run. The repeated stratified design preserves approximate class frequencies in each fold and holds out different observations across repeats. However, samples collected from the same experimental group may violate independence assumptions; the historical dataset metadata do not support a stronger group-aware split. Neither high fold means nor small fold dispersions warrant certainty about future deployment conditions.

For Iris the out-of-fold errors arise only between versicolor and virginica in this run. That is an empirical description, not evidence that certain individual flowers were incorrectly measured or that the taxonomy is ambiguous. Determining the reason for each error would require inspecting original field records and independently validating labels. Similarly, Wine errors should not automatically be attributed to faulty chemistry or mislabelling. A more complete error audit would preserve fold-level probabilities, identify high-uncertainty cases, examine calibrated decision scores and inspect subgroup performance where justified by data provenance and sample sizes.

16 Conclusion

This self-contained study ran a transparent LDA comparison on the traditional Iris dataset and a separate Wine dataset. In the stated repeated cross-validation protocol, LDA achieved 97.87% average accuracy on Iris and 98.93% on Wine, substantially exceeding the respective majority-class baselines. The package includes runnable code and raw JSON results so that the reported scores can be checked. A separate independent numerical reconstruction also matches Fisher's published binary discriminant coefficients and within-group matrix to the precision expected from printing and rounding. This does not imply that historical source code exists or that modern cross-validation scores were reported by Fisher. The new WDBC CSV is curated independently from Iris/Wine, although the observations were collected previously by the dataset owners. The Wine stress test is derived, not independently collected. The LLM judge exercise supports repeatable scoring of archived decisions rather than repeatable model generation. It is a verifiable historical numerical replication plus independent empirical case study; the student must still check assignment requirements beyond these experiments. A complete course project would need to address those gaps explicitly rather than hiding them.

AI Assistance Disclosure

Generative AI assisted the preparation of text, code, report structure, and the archived ten-claim factual judgement exercise. The reported numerical measurements were computed by executing the accompanying Python script. The submitting student remains responsible for verifying the sources, code, analysis, and compliance with the university's declaration rules.

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